1. Pallets

1.1 Technical characteristic

The pallets are used to transport the elements on which the G05 machine is to carry out tasks. Four different formats of pallets exist and they are classified according to their length (overall):

- 60 x 60 mm

Dimension	**Value**
Length	~ 60 mm
Width	60 mm
Height *(see note)*	37 mm
Weight	~ 0.14 kg.

- 120 x 120 mm

Dimension	**Value**
Length	~ 120 mm
Width	120 mm
Height *(see note)*	37 mm
Weight	~ 0.6 kg.

- 160 x 160 mm

Dimension	**Value**
Length	~ 160 mm
Width	160 mm
Height *(see note)*	37 mm
Weight	~ 1.25 kg.

- 240 x 240 mm

Dimension	**Value**
Length	~ 240 mm
Width	240 mm
Height *(see note)*	37 mm
Weight	~ 2.8 kg.

INFO
The height of the base may vary according to the requirements of each application.

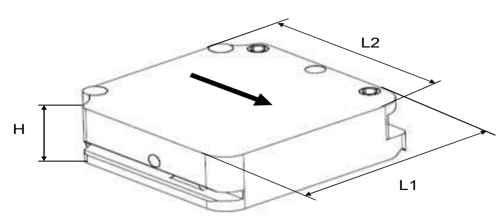

Figure 441-1: Designation of the pallet dimensions.

1.2 Description of standard

The base of the pallet is formed from one piece of synthetic material (polyamide). The pallet has the following characteristics:

(1) V-shaped groove.
Serves as a longitudinal guide when the pallet moves in the machine and its precise position in an axis parallel to its movement.

(2) Dampers.

(3) Patch for presence sensor.
This metal patch allows the pallet to be detected by an inductive presence sensor at the entrance to the machine.

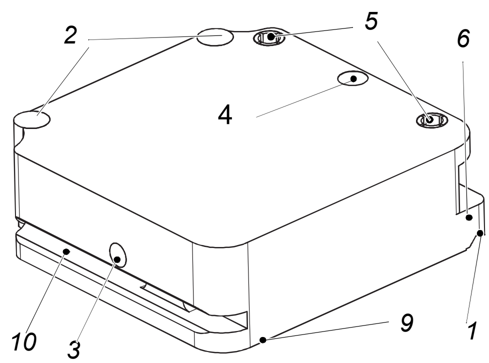

Figure 441-2: 120 mm pallet (view from the front).

(4) Number tag.
Each pallet possesses a unique number written to an electronic label (tag).

(5) Transfer guidebushes.
The pallet advances by means of pins which enter the bushes.

(6) Stop surfaces.
Upon input to the machine, the pallet is held by this face, it ensures its position during take-up by the transfer comb.

(7) Locking pins.
A system of taper pins straddling the positioning pin allows accurate positioning of the pallet.

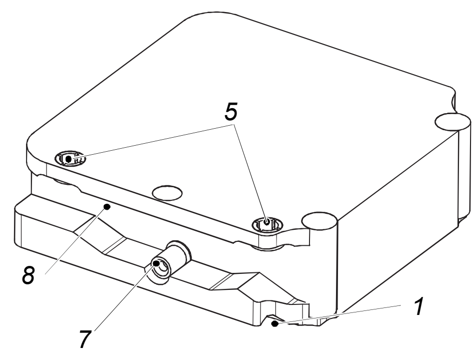

Figure 441-3: 120 m pallet (rear view).

(8) Groove for positioning pin and feed system.
On the one hand, the groove allows the system of combs and taper pins to move. On the other hand, a projecting pin in the entry system makes it possible to lock in position any badly positioned pallet.

(9) Vertical bearing surface.
An element situated on the lower face used for positioning it on the vertical axis.

(10) Front slot.
The slot is used on the machine to prevent the pallet from lifting during transfer. This slot is also used during management of the pallets on conveyor belts using stoppers.

i **INFO**
The accuracy of positioning is guaranteed by the following elements:
(1) V-shaped groove.
(7) Locking pin.
(9) Bearing surface.
It is vital that these surfaces should remain clear and clean in order to guarantee the accuracy of locking of the pallet on the machine.

1.3 Description of different V6 pallets

1.3.1 Pallet 60 x 60 (V6)

(1) Locking pins (1x).
(2) Transfer guide bush (2x).
(3) Dampers (2x).
(4) Presence control (1x).
(5) Slot for stopper.
(6) RFID tag (1x).

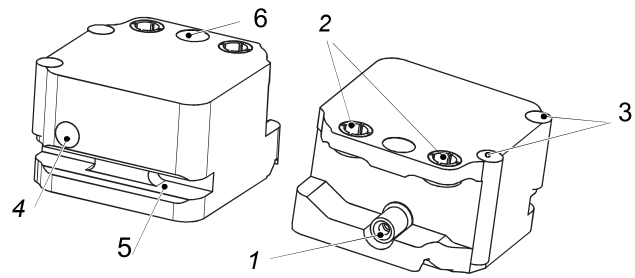

Figure 441-4: Description 60 mm pallet - V6.

1.3.2 Pallet 120 x 120 (V6)

(1) Locking pins (1x).
(2) Transfer guide bush (2x).
(3) Dampers (2x).
(4) Presence control (1x).
(5) Slot for stopper.
(6)

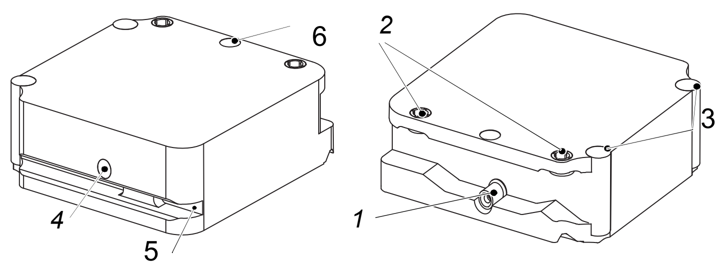

Figure 441-5: Description 120 mm pallet V6.

1.3.3 Pallet 160 x 160 (V6)

(1) Locking pins (2x).
(2) Transfer guide bush (2x).
(3) Dampers (2x).
(4) Presence control (1x).
(5) Slot for stopper.
(6) RFID tag (1x).
(7) Transfer plate.

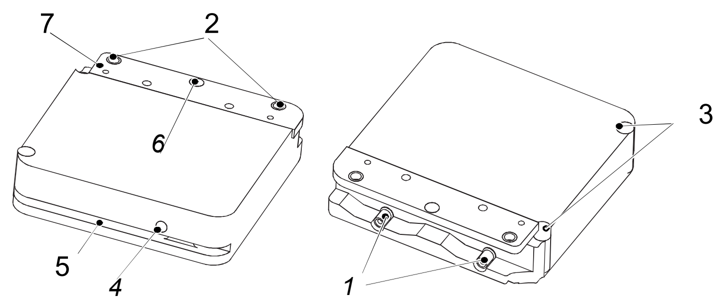

Figure 441-6: Description 160 mm pallet - V6.

1.3.4 Pallet 240 x 240 (V6)

(1) Locking pins (2x).
(2) Transfer guide bush (3x).
(3) Dampers (2x).
(4) Presence control (1x).
(5) RFID tag (1x).
(6) Transfer plate.

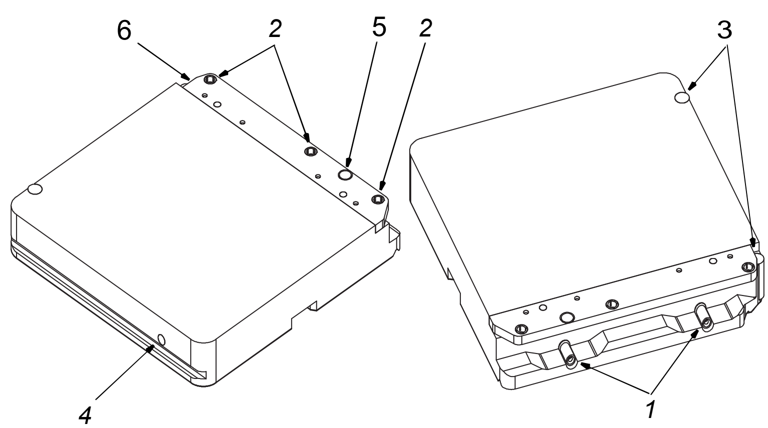

Figure 441-7: Description 240 mm pallet - V6.

1.4 Description of different V5 pallets

INFO: Usually used for multiple pallet transfer per stroke.

1.4.1 Pallet 60 x 60 (V5)

(1) Locking pins (1x).
(2) Transfer guide bush (2x).
(3) Dampers (2x).
(4) Presence control (1x).
(5) Slot for stopper.
(6) RFID tag (1x).

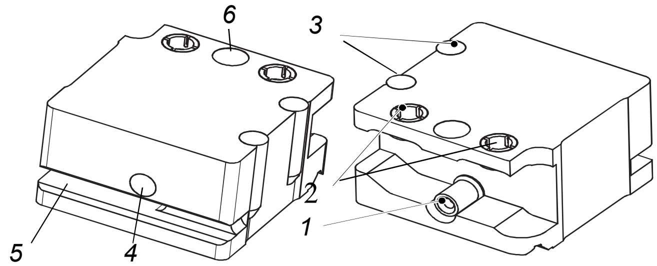

Figure 441-8: Description 60 mm pallet - V5.

1.4.2 Pallet 120 x 120 (V5)

(1) Locking pins (1x).
(2) Transfer guide bush (2x).
(3) Dampers (2x).
(4) Presence control (1x).
(5) Slot for stopper.
(6) RFID tag (1x).

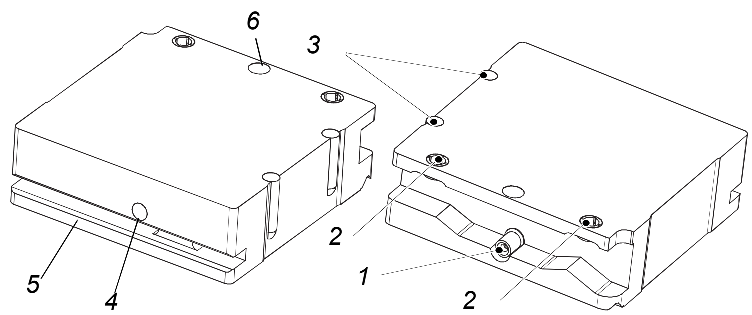

Figure 441-9: Description 120 mm pallet - V5.

1.4.3 Pallet 160 x 160 (V5)

(1) Locking pins (2x).
(2) Transfer guide bush (2x).
(3) Dampers (2x).
(4) Presence control (1x).
(5) Slot for stopper.
(6) RFID tag (1x).
(7) Transfer plate.

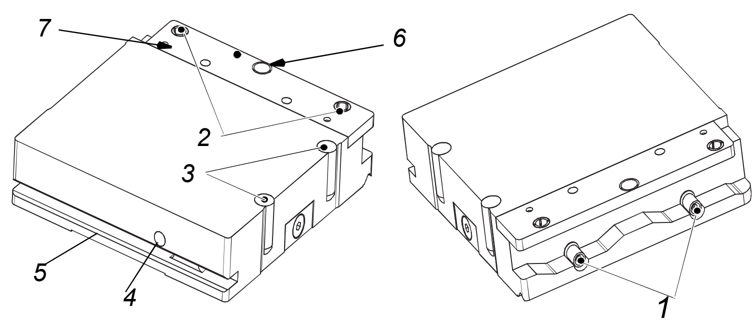

Figure 441-10: Description 160 mm pallet - V5.

1.5 Checks

INFO: Replace the defective pallets. A pallet in poor condition can be a source of operating problems and/or a reduction in production quality.

1.5.1 Visual inspection of the V slot and the underside of the pallets

The lower face of the pallets may be damaged due to friction on the guide. For example, if the locking is not carried out correctly.

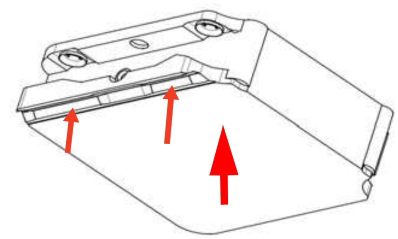

Figure 441-11: Visual inspection of the underside of the pallets.

1.5.2 Visual inspection of the mobile stop area

The support surface may be deteriorated in contact with the movable stop.

INFO: Damage of this tiny surface can alter the pallet pick-up of the machine.

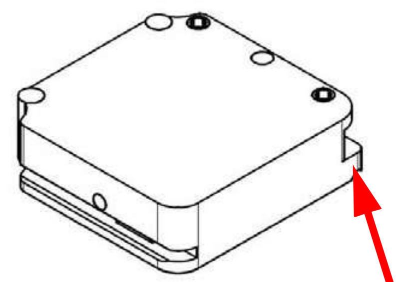

Figure 441-12: Contrôle visuel de la surface d'appuie de la butée.

1.5.3 Visual inspection of the locking pin

The locking pin may be deteriorated, for instance, if one of the locking pins has been deformed.

INFO: Deformation of the pin can cause insufficient locking accuracy.

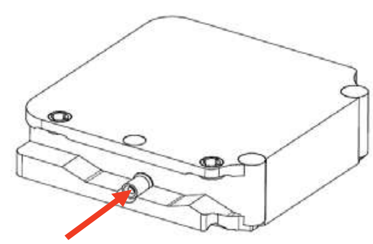

Figure 441-13: Visual inspection of the locking pin.

1.5.4 Visual inspection of the transfer guidebushes

The transfer guide bush may be deteriorated. For instance, this can happen if one of the land pins has been deformed.

INFO: Deformation of the transfer guide bush can result in inaccurate movements.

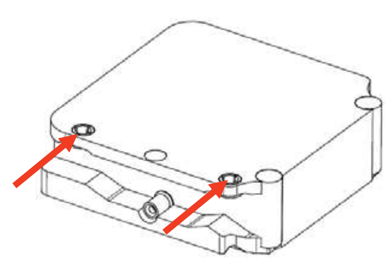

Figure 441-14: Visual inspection of the locking pin receivers.

1.6 Cleaning

1.6.1 Pallet cleaning

The pallets must be cleaned periodically, and the metallic parts (locking pin and transfer bushes) must be parsimoniously lubricated.

See "910-9.7 Periodic maintenance" on page 9-15.

INFO: Only the metallic parts are to be greased!

INFO: If the pallets are de-greased, it is mandatory to re-apply grease to the metallic parts sparingly.

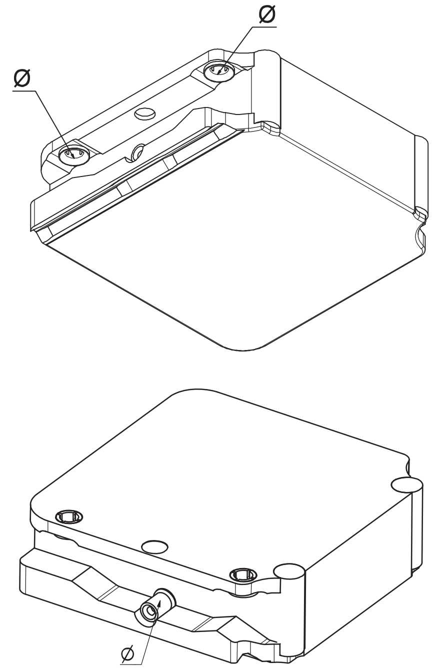

Figure 441-15: Pallet cleaning/lubricating.